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ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

Main Supervisor
Mr. Samadhi Rathnayake
Lecturer
Faculty of Computing SLIIT

Co-Supervisor
Mr. Amila Nuwan Alexander
Assistant Lecturer
Faculty of Computing SLIIT

Problem Statement and Motivation

Children with ASD and Down Syndrome often experience challenges with:

- Attention control
- Object recognition
- Interaction with learning tools
- Understanding decision-making processes

Most existing learning tools rely on:

- Touch interaction
- Teacher supervision
- Limited personalization

Stakeholders

Primary Users

- Children of SERD
- SERD Teachers

Secondary Users

- Special education teachers
- Therapists
- Parents

Collaborating Institution

- SERD – Special Education Resource Development School Kelaniya
- SERD provides support and educational resources for children with special needs and helped us understand real learning challenges and requirements.

Observed Issue

- Learning activities with robots are often slow and not interactive.
- Manual sorting or pattern recognition is inconsistent.
- Existing systems lack real-time, understandable feedback for children.

Research Gap

Existing assistive learning technologies have several limitations:

- Lack of natural interaction methods
- Limited explainability of AI decisions
- Heavy reliance on wearable devices or manual input
- Limited integration between vision, gaze interaction, and AI guidance

There is a need for an adaptive, explainable AI system that supports natural interaction for children with special needs.

Research Objective

- Develop an AI-powered interactive learning system using gaze interaction and computer vision
- Provide explainable AI feedback for clear guidance
- Increase engagement of children in learning activities
- Assist teachers with real-time insights and AI explanations
- Assess system effectiveness on learning outcomes and usability

Solution Overview

We propose an Adaptive Explainable AI Learning System that enables children to interact with pattern identification object texture recognition and gaze-based interaction.

Computer Vision for object texture recognition

Pattern Recognition to analyze behavior

Gaze Tracking to understand attention

Explainable AI to explain decisions

SYSTEM ARCHITECTURE

Texture Recognition

- Sorting materials like glass, plastic, rubber, and metal is important in many industries.
- Current systems cannot sort these materials accurately in real time.
- This project introduces a smart system using vision and sensors to identify materials.
- The system supports real-time detection.
- It also allows human feedback for future improvements.

Gaze Based Intent Recognition

- Hands-free interaction without wearable hardware
- Uses gaze detection for safe robot activation
- Appearance-based gaze classification with region mapping
- Workflow: Face Detection → Gaze Classification → Robot Trigger
- Designed for reliable real-time deployment

Pattern Recognition

- Object pattern sorting is inefficient and often manual.
- Existing systems lack human-robot collaboration.
- Uses computer vision for intelligent pattern recognition.
- NLP-based voice recognition enables voice commands.
- Supports real-time object detection and robotic sorting.
- Improves efficiency and human-robot interaction.

Smart Recommendation

- Manual object-based decision making is slow and inconsistent.
- Introduces an object recommendation system for real-time decisions.
- Provides high accuracy with fast performance.
- Uses Explainable AI for transparent and understandable decisions.

System Integration

Non Functional Focus

- Lack of natural interaction methods
- Limited explainability of AI decisions
- Heavy reliance on wearable devices or manual input
- Limited integration between vision, gaze interaction, and AI guidance

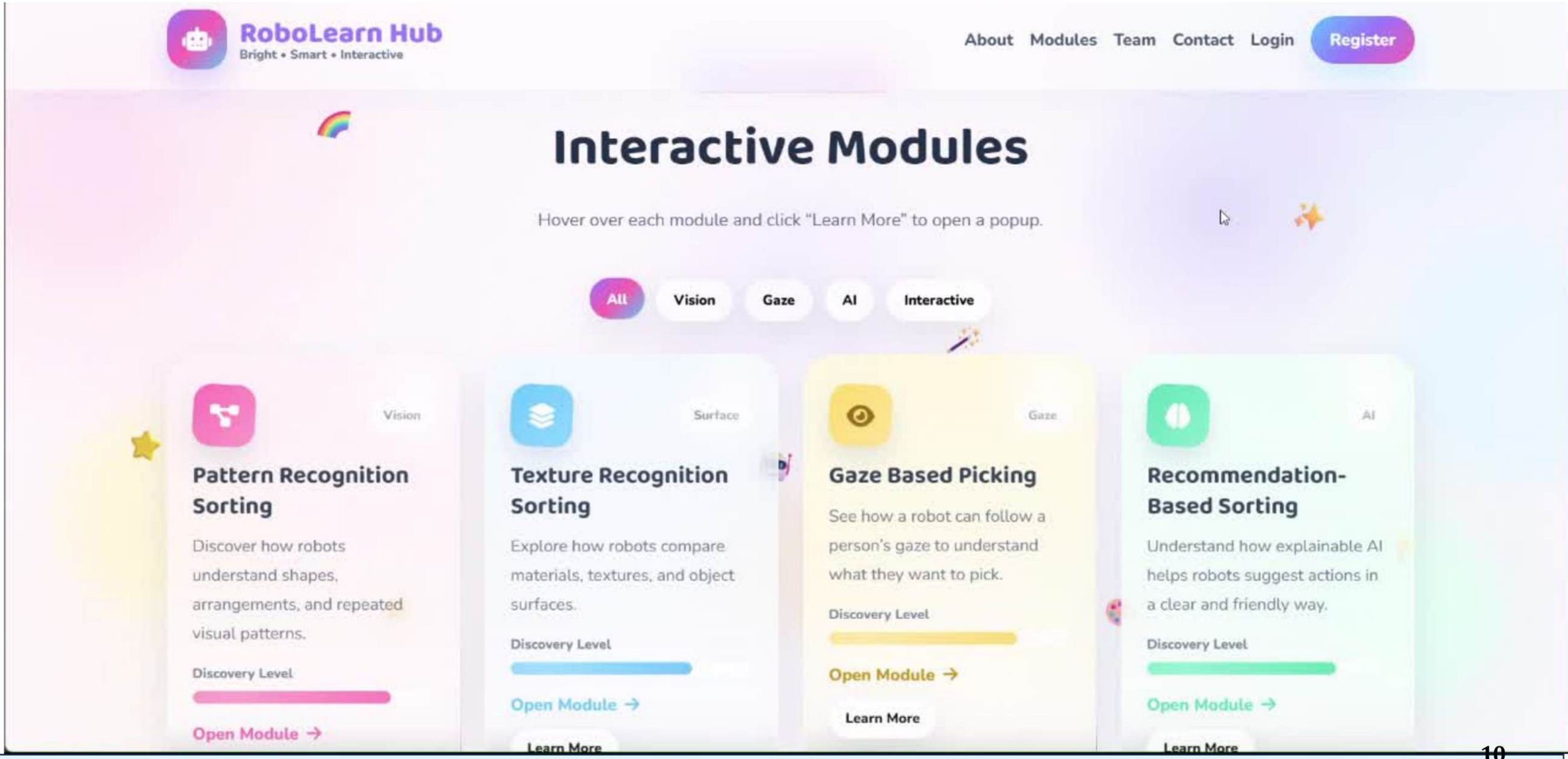
Cost & Feasibility

Prototype components include:
Estimated prototype cost:
$ 155 - $ 160

Current Completion Status

92% Completed

- System Safty Check
- Quality Check with Stake Holders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Pattern Recognition Sorting
Discover how robots understand shapes, arrangements, and repeated visual patterns.
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Texture Recognition Sorting
Explore how robots compare materials, textures, and object surfaces.
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Gaze
Gaze Based Picking
See how a robot can follow a person's gaze to understand what they want to pick.
Discovery Level
Open Module →
Learn More
AI
Recommendation-Based Sorting
Understand how explainable AI helps robots suggest actions in a clear and friendly way.
Discovery Level
Open Module →
Learn More

EXPLORER

live_sorting_esp32.py | 3_main_runtime_ml.py | live_sorting.py | yesh.py Dataset | yesh.py nlp_yolo_project

nlp_yolo_project > live_sorting.py > ...

```
194             system_state = ROBOT_RECORDING
195             robot_pattern = []
196             instruction_msg = "Robot recording... | Press L again when finished"
197         elif system_state == ROBOT_RECORDING and l_pressed and not prev_l_state:
198             system_state = ROBOT_DONE
199             instruction_msg = "Robot pattern recorded | Press P to record human pick | Press P again when done"
200
201         prev_l_state = l_pressed
202
203         if system_state == ROBOT_RECORDING:
204             for obj in detected_objects:
205                 if len(robot_pattern) < len(pattern_to_show) - 1:
206                     expected = pattern_to_show[len(robot_pattern)]
207                     if obj == expected:
208                         robot_pattern.append(obj)
209
210         # ==============================
211         # Human Recording
212         # ==============================
```

PATTERN_RECOGNITION

- Dataset
  - yesh.py
- Frames
- nlp_yolo_project
  - __pycache__
    - xai_module.cpython-311.pyc
  - 1_create_dataset.py
  - 2_train_nlp_model.py
  - 3_main_runtime_ml.py
  - best.pt
  - live_sorting.py
  - nlp_dataset.json
  - nlp_lr_model.pkl
  - nlp_vectorizer.pkl
  - xai_module.py
  - yesh.py
- pattern_dataset
- venvpattern
  - Include
  - Lib
  - Scripts
  - share
  - pyvenv.cfg
- YolotxtData

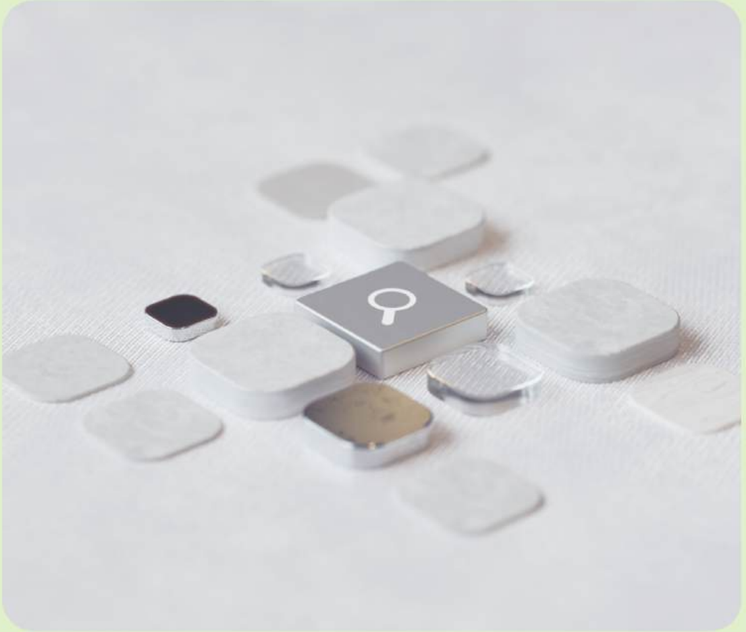
OUTLINE

TIMELINE

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

python - nlp_yolo_project

```
(venvpattern) PS D:\4th year\Research\Pattern_Recognition\nlp_yolo_project> python 3_main_runtime_ml.py
Traceback (most recent call last):
  File "D:\4th year\Research\Pattern_Recognition\nlp_yolo_project\3_main_runtime_ml.py", line 118, in <module>
    ret, frame = cap.read()
                 ^^^^^^^^^^
KeyboardInterrupt
(venvpattern) PS D:\4th year\Research\Pattern_Recognition\nlp_yolo_project> python live_sorting.py
✅ YOLO model loaded
✅ Camera connected
```

0 ⚠ 0 | Ln 175, Col 51 | Spaces: 4 | UTF-8 | CRLF | { } Python | Signed out | 3.11.9 (venvpattern) | venvpattern (3.11.9) | Go Live

THANK YOU!